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# Decomposing Error in Continual World Models

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## Abstract

A continual world model is evaluated after each update, but post update accuracy does not indicate whether the update itself tracked a change in the environment. Endpoint error decomposes into inherited error and update mismatch through the identity  $\varepsilon^1 = \varepsilon^0 + (\delta_M - \delta_E)$ , where  $\delta_M$  and  $\delta_E$  are model and environment version prediction differences on the same interventional query. Endpoint evaluation can favor replacement over faithful revision, while mismatch alone rewards preserving inherited bias. Controlled linear experiments reverse the ranking of update rules on all 544 usable trials at one step. A learned neural model on a versioned nonlinear simulator reproduces the predicted direction: across a base-data sweep, reversals fall from 95% to 6% as inherited error falls from 0.038 to 0.004. We propose paired reporting of endpoint, inherited, and mismatch errors over a declared query panel and multiple horizons as an evaluation protocol. Code is available here.

## 1 Motivation

Continual world modelling concerns sequences of environments  $E^0, E^1, \dots$  and models  $M^0, M^1, \dots$ , connected by an update operator. Existing evaluations ask whether  $M^1$  predicts  $E^1$  accurately. Recent work also measures backward and forward transfer, and tests changed and stable knowledge separately [Harrington et al., 2026]. These are post-hoc assessments of the learner as they do not evaluate the update.

A learner that discards a biased  $M^0$  and refits on abundant post-change data may achieve low endpoint error while correcting old errors that the environment did not change. Conversely, a selective updater that revises exactly the changed mechanism and retains existing errors yields high endpoint error despite faithful tracking of the environmental change. Neither behavior is universally preferable. Replacement may be appropriate when data is abundant, while selective revision may be required for retention as endpoint error alone cannot distinguish them.

## 2 Decomposition of update error

Fix a version-comparable query  $z = (q, i, h)$ : predictive target  $q$ , intervention  $i$ , and horizon  $h$ . Let  $P_E^v(z)$  and  $P_M^v(z)$  denote environment and model predictions at version  $v \in \{0, 1\}$ . Define

$$\varepsilon^v(z) := P_M^v(z) - P_E^v(z), \quad \delta_E(z) := P_E^1(z) - P_E^0(z), \quad \delta_M(z) := P_M^1(z) - P_M^0(z).$$

Pointwise in  $z$ ,

$$\underbrace{\varepsilon^1}_{\text{endpoint error}} = \underbrace{\varepsilon^0}_{\text{inherited error}} + \underbrace{(\delta_M - \delta_E)}_{\text{update mismatch}}. \quad (1)$$

This identity holds by definition but the norms do not add because  $\|\varepsilon^1\| \neq \|\varepsilon^0\| + \|\delta_M - \delta_E\|$  except when the terms are positively collinear. The sign of each term determines whether mismatch amplifies or cancels inherited error. Therefore, endpoint and inherited norms alone are insufficient.

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Equation 1 also fixes what mismatch is. Rearranged,  $\delta_M - \delta_E = \varepsilon^1 - \varepsilon^0$ , so mismatch is the change in signed prediction error across the update and requires no measurement beyond the endpoint and inherited errors it is computed from. Its additional cost is temporal rather than causal as the panel must be fixed and evaluated at version 0 before the shift rather than reconstructed afterwards. The decomposition therefore contributes a reporting protocol rather than a new observable and the open question is the regime in which the two views disagree.

Minimizing  $\mathbb{E}_z \|\delta_M - \delta_E\|$  favors  $\delta_M = \delta_E$ , hence

$$P_M^1 = P_M^0 + (P_E^1 - P_E^0) = P_E^1 + \varepsilon^0.$$

The minimizer preserves inherited bias:  $\varepsilon^1 = \varepsilon^0$ . A model that instead achieves  $\varepsilon^1 = 0$  incurs mismatch  $-\varepsilon^0$ . Therefore, mismatch penalizes any correction to inherited error even when that correction improves endpoint accuracy. Because mismatch measures only the agreement between two changes, minimizing it alone guarantees that inherited bias persists. However, endpoint error has the complementary limitation as it rewards corrections but cannot distinguish faithful revisions from wholesale replacement. Because neither metric suffices alone, we report endpoint, inherited, and mismatch error jointly. The same decomposition extends to sequences of updates. Across  $K$  updates, the signed identity telescopes:

$$\varepsilon^K = \varepsilon^0 + \sum_{k=1}^K (\delta_M^k - \delta_E^k).$$

Each term in the sum carries a sign, so mismatches across rounds can accumulate or offset one another. A single round with low mismatch therefore does not constrain  $\varepsilon^K$  on its own.

### 3 Experimental demonstration

We use a linear structural dynamical system  $x_{t+1} = Ax_t + \eta$  in five dimensions. This choice makes the required environmental counterfactual observable rather than assumed: under  $\text{do}(x_0 = e_i)$ , the horizon- $h$  prediction is  $A^h e_i$  so  $\delta_E(h) = A_1^h - A_0^h$  can be computed exactly. We generate sparse matrices  $A_0$  with spectral radius 0.70 and produce  $A_1$  by changing one transition entry by 0.35. Data is generated by isotropic Gaussian inputs and noise  $\sigma = 0.25$ . We fit a ridge base model ( $\lambda = 40$ ) on 400 pre-change samples and give each update 400 post-change samples.

We compare post-change refitting, 1:1 replay, and an oracle baseline constrained to modify only the changed transition entry. We attempt 600 seeded systems and retain 544 after predeclared stability and propagation checks. All reported quantities are relative Frobenius norms over the five unit interventions:

$$\begin{aligned} \text{ENDPOINT}(h) &= \frac{\|\hat{A}_1^h - A_1^h\|_F}{\|A_1^h\|_F}, & \text{INHERITED}(h) &= \frac{\|\hat{A}_0^h - A_0^h\|_F}{\|A_0^h\|_F}, \\ \text{MISMATCH}(h) &= \frac{\|(\hat{A}_1^h - \hat{A}_0^h) - (A_1^h - A_0^h)\|_F}{\|A_1^h - A_0^h\|_F}. \end{aligned}$$

**Rank reversals.** At  $h = 1$ , refit achieves endpoint/mismatch 0.039/0.470, while the oracle achieves 0.096/0.047. Refit wins on endpoint and the oracle wins on mismatch in all 544 seeds (Wilson 95% interval [99.3, 100.0]). The reversal persists at 93.8% for  $h = 5$  and 66.2% for  $h = 12$ . Equation 1 accounts for this pattern as at  $h = 1$ , the median Frobenius cosine between refit’s inherited error and mismatch is  $-0.929$ , so they cancel substantially. Furthermore, the oracles cosine is  $-0.079$  so its inherited error remains visible at the endpoint.

**Effect of inherited error.** Sweeping ridge strength moves measured one-step inherited error from 0.040 to 0.172 and disagreement from 53.1% to 100%. Each condition contains 544 seeds and the sweep does not estimate a population response curve. The 53.1% disagreement point uses ordinary least squares without regularization, so finite-sample estimation error alone suffices to produce rank reversal. With 8,000 base samples, inherited error falls to 0.009 and disagreement to 0%. The direction of this dose response follows from Equation 1, since refitting on abundant post-change data

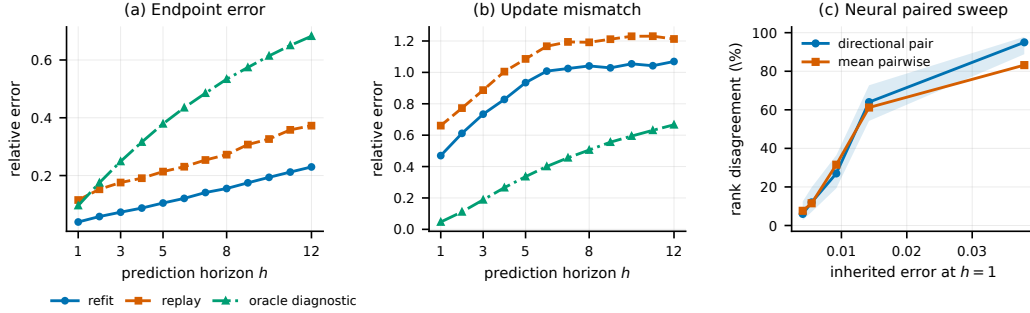


Figure 1: Endpoint and mismatch errors produce opposite rankings in both settings. Linear system (544 seeds): (a) refit yields the lowest endpoint error, while (b) the oracle baseline yields the lowest mismatch. (c) Neural pendulum: pairwise disagreement between refit and proximal updates increases with inherited error. Shading shows Wilson 95% confidence intervals; lines connect the five  $N_0$  conditions.

drives  $\varepsilon^1$  toward zero and therefore drives mismatch toward  $\|\varepsilon^0\|$ , while a conservative update leaves inherited error in place. What the sweep measures is the scale at which the two views separate, namely an inherited error near 0.04, which unregularized least squares on 400 samples already reaches. It does not claim that inherited error is the only possible source of rank reversal.

**Multi-round ordering.** The identity does not make early mismatch predictive of  $\varepsilon^K$ , but it does not forbid it. In a ten round variant with 25 samples per round and one transition entry changing per round, we rank four update rules within each of 400 seeds by their round one scores and compare that ordering against the ordering by endpoint error after the tenth round. At  $h = 5$ , round-one mismatch recovers the final pairwise ordering on 70.8% of pairs against 51.0% for round one endpoint, a paired difference of +0.198 with bootstrap interval [0.173, 0.222], and mean Spearman correlation rises from 0.022 to 0.496. The advantage is small at  $h = 1$  (+0.022, [0.010, 0.035]). Appendix A gives the full table. This is an association within one construction and does not license mismatch as an optimization objective.

**Non-oracle and neural checks.** A non-oracle proximal update in the linear system,

$$\hat{A}_1 = (YX^\top + \mu\hat{A}_0)(XX^\top + \mu I)^{-1}$$

produces one-step reversal rates from 20.4% to 85.8% across seven anchoring strengths. At  $\mu = 80$ , the rate is 467/544 (95% Wilson interval [82.7, 88.5]).

We then train a 2–64–64–2 tanh delta-MLP on a fully observed, versioned discrete-time damped pendulum simulator (RK4,  $dt = 0.05$ ,  $g : 9.8 \rightarrow 11.5$ ). Exact simulator rollouts define  $\delta_E$  on a fixed 18 state panel at  $h \in \{1, 3, 5, 10, 20\}$ . For each of 100 paired seeds, five base data conditions use nested prefixes and share the same post-change data and validation holds out complete trajectories. We compare refit, 1:1 replay, and five parameter anchored proximal updates, retaining all 500 runs.

Figure 1c shows that both the directional reversal rate and mean pairwise rank disagreement increase with measured inherited error. Equivalently, both decrease as larger base datasets reduce inherited error. In the highest error condition, refit and the most anchored update reverse order on 95/100 seeds at  $h = 1$  and 81/100 at  $h = 20$ . From the smallest to largest base-data condition, pairwise disagreement decreases on 99/100 paired seeds. The cancellation cosine also moves toward zero with stronger anchoring. The reversal pattern persists under learned nonlinear rollout.

## 4 Proposed reporting protocol

We propose the following requirements for evaluating continual world model updates. These requirements address the measurement of error components, the documentation of environmental changes, the handling of edge cases in aggregation, and the scope of empirical validation.

1. **Error decomposition.** Report endpoint, inherited, and mismatch error on the same pre-declared query panel. Predeclaration prevents adaptive selection of queries that favor one update rule over another. Retain signed predictions rather than absolute errors whenever possible, because the identity in Equation 1 holds for signed quantities and cancellation patterns are visible only in the signed regime.
2. **Environment delta documentation.** State the source of the environment delta explicitly. Acceptable sources include paired versions of a simulator, outputs from controlled system identification procedures, or a benchmark oracle. The source determines whether  $\delta_E$  is exact or estimated, which constrains the interpretation of mismatch values.
3. **Query specification.** Declare the query support, the weighting scheme over queries, the norm used for aggregation, and the convention for handling queries where the true effect is zero. Zero effect queries are particularly important because they test whether the update introduces spurious changes in mechanisms that the environment left unchanged.
4. **Temporal scope.** Report results at multiple planning horizons, since the relative importance of inherited error and mismatch varies with horizon length. For claims about continual learning over sequences of updates, report multiple update rounds with complete accounting of seeds and any rejected runs.

The query panel must have shared semantics across versions. Changes in decoder, coordinate system, target, or support between versions render subtraction scientifically meaningless even when it remains numerically defined. The protocol enforces comparability through the choice of panel, independently of the optimizer’s causal structure.

Aggregation requires explicit conventions. The maximum of per-query relative errors is undefined when  $\|\delta_E(i)\| = 0$ , which is the subset of queries where unintended changes should be detected. A sensitivity check retaining these queries and comparing a worst-column matrix norm against the global Frobenius ratio showed no additional long horizon separation. The protocol therefore requires an explicit zero effect convention instead of a universal worst case scalar.

## 5 Related Works

Harrington et al. treat arbitrary update operators as the object of a sequential continual learning protocol and test changed facts alongside stable facts [Harrington et al., 2026]. Their observables are post stage accuracy, F1, and transfer. Our decomposition isolates the change in prediction error that those state based scores can hide. LoCA regret measures policy-level adaptation speed after a local environmental change and demonstrates that ordinary task performance can conflate distinct capabilities [van Seijen et al., 2020] as it does not compare model and environment prediction deltas.

Knowledge-editing evaluation studies propagation to entailed facts and preservation of unrelated facts [Cohen et al., 2024], including their coupled optimization [Huang et al., 2026] but we do not claim those criteria as our contribution. Our focus is the model’s and environment’s version deltas in recursively composed dynamics. Counterfactual mechanics benchmarks provide related static, intervention-grounded evaluation [Ness et al., 2026]. Multi-step model error and the need for short model rollouts are established in model based RL [Janner et al., 2019]; we use horizon as an axis of update evaluation rather than rediscovering compounding error of a fixed model.

## 6 Limitations and conclusion

The protocol needs environment predictions on the declared panel at both versions, which a versioned simulator or controlled benchmark supplies exactly. In a deployed setting the constraint is temporal rather than causal as the panel must be evaluated before the shift, since the earlier environment cannot be queried afterwards. Our demonstrations are low-dimensional and fully observed, and the neural extension shows that the effect survives learned nonlinear rollout rather than establishing its prevalence in deep latent world models. Because the  $N_0$  sweep moves the base training set along with inherited error, the paired design identifies an association.

Endpoint error measures whether the model is accurate after the update and mismatch measures whether its revision tracked the environment’s. Reporting both alongside inherited error on a declared panel makes continual evaluation describe the update and not only its endpoint.

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## A Multi-round ordering experiment

The sequential variant uses the same linear system as Section 3. A ridge base model ( $\lambda = 40$ ) is fit on 400 pre change samples, after which the world performs ten successive updates. Each round changes one uniformly chosen transition entry by  $\pm 0.25$ , subject to a spectral guard  $\rho < 0.95$ , and supplies 25 fresh samples. The per-round sample budget is deliberately scarce, since with abundant data per round post change refitting is already well determined and the sequential setting is not continual in any useful sense. We compare four rules: post change refitting, replay on the accumulated buffer, a proximal update anchored to the previous model at  $\mu = 30$ , and the oracle diagnostic restricted to the entry that changed. Results are over 400 seeds.

Within each seed we rank the four rules by a round one score and compare that ranking against the ranking by endpoint error after the tenth round. Table 1 reports the fraction of the six rule pairs ordered correctly, the fraction of seeds whose best rule is identified, and the mean Spearman correlation between the two rankings.

| Round-one predictor       | $h = 1$  |      |          | $h = 5$  |      |          |
|---------------------------|----------|------|----------|----------|------|----------|
|                           | pairwise | best | Spearman | pairwise | best | Spearman |
| Endpoint error            | 0.755    | 0.64 | 0.617    | 0.510    | 0.26 | 0.022    |
| Update mismatch           | 0.777    | 0.64 | 0.675    | 0.708    | 0.48 | 0.496    |
| Mean mismatch over rounds | 0.740    | 0.64 | 0.619    | 0.596    | 0.47 | 0.236    |

Table 1: Recovery of the ten round endpoint ordering of four update rules from round one scores, over 400 seeds. Paired seed bootstrap with 10,000 resamples, mismatch minus endpoint: pairwise  $+0.022$   $[0.010, 0.035]$  and Spearman  $+0.058$   $[0.034, 0.084]$  at  $h = 1$ ; pairwise  $+0.198$   $[0.173, 0.222]$  and Spearman  $+0.474$   $[0.416, 0.531]$  at  $h = 5$ .

Round one mismatch orders the rules better than round one endpoint at both horizons, and the gap widens at the planning horizon. Three qualifications apply. The oracle rule has both the lowest round one mismatch and the lowest ten round endpoint error, so part of the advantage comes from ranking it correctly. The statistics are computed per seed over all six rule pairs rather than on the oracle alone, but the result should not be read as independent of it. Neither predictor recovers the final ordering exactly, so this is added ordering information rather than a forecast. The measurement is confined to this construction, and it doesn’t make mismatch safe to minimize directly for the reason given in Section 2.

A separate observation from the same runs supports multiround reporting independently of the mismatch metric. Replay moves from third of the four rules at round one to last by round ten, with median endpoint error rising from 0.152 to 0.357 at  $h = 1$  because its buffer accumulates data from environments that no longer exist.